



**HOSTEL MESS QUALITY ASSESSMENT
AND ANALYTICS SYSTEM
PROJECT REPORT**

**Submitted in partial fulfillment of the requirements
for the course**

PYTHON PROGRAMMING

Submitted By

Submitted To

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HOSTEL MESS QUALITY ASSESSMENT AND ANALYTICS SYSTEM

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1. INTRODUCTION

The Hostel Mess Quality Assessment and Analytics System is a Python-based application designed to improve the quality of food services provided in hostel mess facilities. The system provides a structured platform through which students can submit feedback regarding food quality, register complaints, record food wastage information, and vote for preferred dishes.

The collected information is processed and analyzed to generate meaningful reports that can assist hostel management in making informed decisions. The system aims to improve food quality, increase student satisfaction, reduce food wastage, and establish an organized mechanism for performance evaluation.

2. PROBLEM STATEMENT

Hostel students frequently experience issues related to food quality, hygiene, quantity, variety, and food wastage. Traditional feedback mechanisms are often informal and unstructured, making it difficult for hostel administrators to understand student concerns and identify recurring problems.

The absence of a centralized system results in ineffective communication between students and management. Therefore, there is a need for a system that can systematically collect student feedback, analyze complaints, monitor food wastage, and provide actionable insights for improving mess services.

The Hostel Mess Quality Assessment and Analytics System addresses these challenges by providing a platform for feedback collection, complaint tracking, waste analysis, and report generation.

3. OBJECTIVES

The primary objectives of the project are:

- To collect structured feedback regarding food quality.
 - To monitor complaints related to mess services.
 - To analyze food wastage patterns.
 - To identify the most preferred dishes through student voting.
 - To generate analytical reports for decision-making.
 - To improve overall student satisfaction.
 - To promote efficient utilization of mess resources.
 - To support continuous service improvement through data-driven analysis.
-

4. FUNCTIONAL REQUIREMENTS

The system provides the following functionalities:

4.1 Food Feedback Management

- Students can submit feedback regarding meals.
- Students can rate taste, quantity, hygiene, and presentation.
- Students can provide recommendations and comments.

4.2 Complaint Management

- Students can register complaints.
- Complaints can be categorized as:
 - Taste Issues
 - Hygiene Issues
 - Quantity Issues
 - Variety Issues

4.3 Waste Monitoring

- Food wastage records can be entered.
- Waste information can be analyzed and reported.

4.4 Voting System

- Students can vote for preferred dishes.
- Vote counts are maintained and analyzed.

4.5 Analytics Generation

- Generate detailed reports.
- Calculate satisfaction metrics.
- Display overall mess performance status.

5. NON-FUNCTIONAL REQUIREMENTS

Reliability

The system should consistently produce accurate results.

Usability

The system should provide an easy-to-use menu-driven interface.

Maintainability

The project should be modular and easy to modify.

Performance

The system should respond quickly to user inputs.

Scalability

Additional features can be incorporated in future versions.

6. SYSTEM ARCHITECTURE

SYSTEM ARCHITECT DIAGRAM



Architecture Description

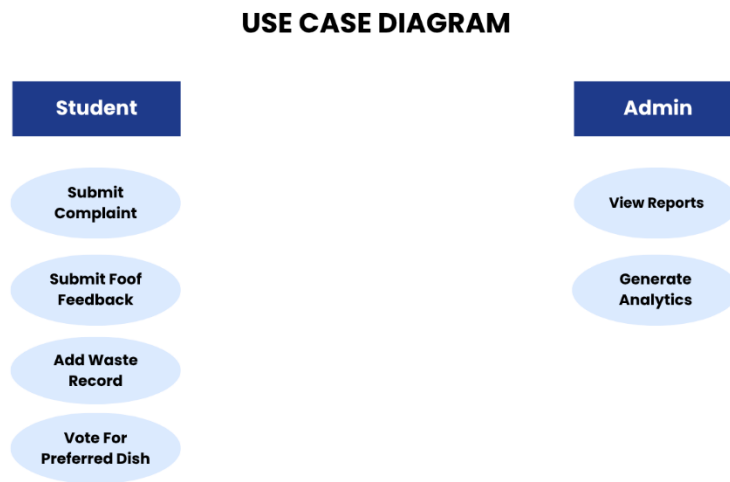
The system follows a modular architecture.

The Student Interface collects feedback, complaints, waste information, and voting data. The collected information is processed through individual modules such as the Feedback Module, Complaint Module, Waste Monitoring Module, and Voting Module.

The Analytics Module processes the collected information and generates meaningful reports that can be used by hostel administrators to evaluate mess performance and take corrective actions.

7. DESIGN DIAGRAMS

7.1 Use Case Diagram



Description

The Use Case Diagram illustrates interactions between users and the system.

Students can:

- Submit Feedback
- Submit Complaints
- Add Waste Records
- Vote for Preferred Dishes

Administrators can:

- View Reports
 - Generate Analytics
-

7.2 Workflow Diagram

WORKFLOW DIAGRAM



Description

The workflow begins with data collection. Students submit feedback, complaints, waste records, and votes. The information is stored and processed by the system. Analytical reports are generated and displayed for decision-making.

7.3 System Architecture Diagram

SYSTEM ARCHITECT DIAGRAM



Description

The architecture demonstrates the flow of information from users through processing modules and finally to report generation.

8. DESIGN DECISIONS AND RATIONALE

Several design decisions were made during the development of the project:

- Python was selected due to its simplicity and ease of implementation.
- A menu-driven console interface was used to ensure usability for all users.
- Modular programming was adopted to improve maintainability.
- Lists and dictionaries were utilized for efficient data handling.
- Analytics reports were included to provide meaningful insights instead of simple data storage.

These design decisions helped create a simple, efficient, and extensible system.

9. IMPLEMENTATION DETAILS

The project was implemented using Python.

The system consists of multiple modules:

feedback.py

Handles student food quality feedback.

complaints.py

Handles complaint registration and complaint analysis.

waste.py

Manages food wastage records.

voting.py

Records voting information and calculates dish popularity.

analytics.py

Processes all collected data and generates reports.

main.py

Acts as the entry point and provides the menu-driven interface.

The application uses Python functions, lists, dictionaries, loops, and conditional statements for implementation.

10. RESULTS

```
===== RESTART: C:\Users\HP\OneDrive\Desktop\Python Sem1\Project\main.py =====  
=====  
HOSTEL MESS QUALITY ASSESSMENT AND ANALYTICS SYSTEM  
=====  
1. Submit Food Feedback  
   2. View Feedback Report  
   3. Submit Complaint  
   4. View Complaint Report  
   5. Add Waste Record  
   6. View Waste Report  
   7. Vote For Preferred Dish  
   8. View Voting Report  
   9. Generate Analytics Report  
  10. Exit  
=====  
Enter your choice : |
```

Main Menu

The main menu allows access to all system functionalities.

```

feedback_data = []

def add_feedback():
    print("\n=====")
    print("          FOOD QUALITY FEEDBACK")
    print("=====")

    dish = input("Dish Name: ")

    taste = int(input("Taste Rating (1-5): "))
    quantity = int(input("Quantity Rating (1-5): "))
    hygiene = int(input("Hygiene Rating (1-5): "))
    presentation = int(input("Presentation Rating (1-5): "))

    recommendation = input("Would you like this dish again? (Y/N): ")

    comment = input("Additional Comment: ")

    feedback = {"dish": dish, "taste": taste, "quantity": quantity, "hygiene": hygiene, "presentation": presentation, "recommendation": recommendation, "comment": comment}

    feedback_data.append(feedback)

    print("\nFeedback Submitted Successfully!")

def view_feedback():
    if len(feedback_data) == 0:
        print("\nNo Feedback Available")
        return

    print("\n=====")
    print("          FEEDBACK REPORT")
    print("=====")

    for feedback in feedback_data:
        print("\nDish :", feedback["dish"])
        print("Taste :", feedback["taste"])
        print("Quantity :", feedback["quantity"])
        print("Hygiene :", feedback["hygiene"])
        print("Presentation :", feedback["presentation"])
        print("Recommend Again :", feedback["recommendation"])
        print("Comment :", feedback["comment"])

```

Feedback Module

Students can provide ratings and comments regarding food quality.

```
complaints = []

def add_complaint():

    print("\n=====")
    print("          COMPLAINT PORTAL")
    print("=====")

    print("1. Taste")
    print("2. Hygiene")
    print("3. Quantity")
    print("4. Variety")

    c = int(input("Choose Category: "))

    if c == 1:
        category = "Taste"

    elif c == 2:
        category = "Hygiene"

    elif c == 3:
        category = "Quantity"

    else:
        category = "Variety"

    description = input("Enter Complaint: ")

    complaints.append({"category": category, "description": description})
```

```

elif c == 2:
    category = "Hygiene"

elif c == 3:
    category = "Quantity"

else:
    category = "Variety"

description = input("Enter Complaint: ")

complaints.append({"category": category, "description": description})

print("\nComplaint Registered!")

def view_complaints():

    if len(complaints) == 0:
        print("\nNo Complaints Available")
        return

    print("\n=====")
    print("          COMPLAINT REPORT")
    print("=====")

    for complaint in complaints:

        print("\nCategory :", complaint["category"])
        print("Issue :", complaint["description"])

```

Complaint Module

Students can submit complaints related to mess services.

```

waste_records = []

def add_waste():

    print("\n=====")
    print("          WASTE MONITORING")
    print("=====")

    dish = input("Dish Name: ")
    served = int(input("Students Served: "))
    waste = float(input("Waste Generated (Kg): "))

    waste_records.append({
        "dish": dish,
        "served": served,
        "waste": waste
    })

    print("\nWaste Data Recorded!")

def waste_report():

    if len(waste_records) == 0:
        print("\nNo Waste Records")
        return

    print("\n=====")
    print("          WASTE REPORT")
    print("=====")

def waste_report():

    if len(waste_records) == 0:
        print("\nNo Waste Records")
        return

    print("\n=====")
    print("          WASTE REPORT")
    print("=====")

    for record in waste_records:

        print("\nDish :", record["dish"])
        print("Students Served :", record["served"])
        print("Waste :", record["waste"], "Kg")

```

Waste Monitoring Module

Records and analyzes food wastage information.

```
votes = {}

def vote_dish():

    print("\n=====")
    print("          NEXT WEEK FOOD VOTING")
    print("=====")

    dish = input("Enter Dish Name: ")

    if dish not in votes:
        votes[dish] = 0

    votes[dish] += 1

    print("Vote Recorded Successfully!")

def voting_report():

    print("\n=====")
    print("          VOTING REPORT")
    print("=====")

    if len(votes) == 0:
        print("No Votes Recorded")
        return

    for dish in votes:

        print(dish, ":", votes[dish], "Votes")
```

Voting Module

Allows students to vote for preferred dishes.

```
|from feedback import feedback_data
from complaints import complaints
from waste import waste_records
from voting import votes

def generate_analytics():

    if len(feedback_data) == 0:
        print("No Data Available")
        return

    taste_total = 0
    quantity_total = 0
    hygiene_total = 0
    presentation_total = 0

    recommend_yes = 0

    for feedback in feedback_data:

        taste_total += feedback["taste"]
        quantity_total += feedback["quantity"]
        hygiene_total += feedback["hygiene"]
        presentation_total += feedback["presentation"]

        if feedback["recommendation"].upper() == "Y":
            recommend_yes += 1

    total = len(feedback_data)
```

```
avg_taste = taste_total / total
avg_quantity = quantity_total / total
avg_hygiene = hygiene_total / total
avg_presentation = presentation_total / total

satisfaction_rate = (recommend_yes / total) * 100

print("\n=====")
print("      HOSTEL MESS ANALYTICS REPORT")
print("=====")

print("Total Feedback Entries :", total)

print("Average Taste Score :", round(avg_taste, 2))
print("Average Quantity Score :", round(avg_quantity, 2))
print("Average Hygiene Score :", round(avg_hygiene, 2))
print("Average Presentation Score :", round(avg_presentation, 2))

print("Total Complaints :", len(complaints))
print("Total Waste Records :", len(waste_records))

print("Student Satisfaction Rate :",
      round(satisfaction_rate, 2), "%")

print("Overall Mess Rating :", round(d, 2))

if o >= 4:
    print("Status : EXCELLENT")

elif o >= 3:
    print("Status : GOOD")

else:
    print("Status : NEEDS IMPROVEMENT")

print("=====")
```

Analytics Report

Displays summarized insights and performance indicators.

```
=====
HOSTEL MESS QUALITY ASSESSMENT AND ANALYTICS SYSTEM
=====
```

1. Submit Food Feedback
2. View Feedback Report
3. Submit Complaint
4. View Complaint Report
5. Add Waste Record
6. View Waste Report
7. Vote For Preferred Dish
8. View Voting Report
9. Generate Analytics Report
10. Exit

```
=====
Enter your choice : 1
```

```
=====
FOOD QUALITY FEEDBACK
=====
```

```
Dish Name: Poha
Taste Rating (1-5): 3
Quantity Rating (1-5): 5
Hygiene Rating (1-5): 4
Presentation Rating (1-5): 5
Would you like this dish again? (Y/N): N
Additional Comment: Too plain
```

```
Feedback Submitted Successfully!
```

```
Enter your choice : 2
```

```
=====
FEEDBACK REPORT
=====
```

```
Dish : Poha
Taste : 3
Quantity : 5
Hygiene : 4
Presentation : 5
Recommend Again : N
Comment : Too plain
```

Enter your choice : 3

=====

COMPLAINT PORTAL

=====

1. Taste
2. Hygiene
3. Quantity
4. Variety

Choose Category: 1

Enter Complaint: Sometimes, the food tastes bland and off

Complaint Registered!

Enter your choice : 4

=====

COMPLAINT REPORT

=====

Category : Taste

Issue : Sometimes, the food tastes bland and off

Enter your choice : 5

=====

WASTE MONITORING

=====

Dish Name: Karakulambu

Students Served: 2000

Waste Generated (Kg): 3

Waste Data Recorded!

Enter your choice : 6

=====

WASTE REPORT

=====

Dish : Karakulambu
Students Served : 2000
Waste : 3.0 Kg

Enter your choice : 7

=====

NEXT WEEK FOOD VOTING

=====

Enter Dish Name: Pasta
Vote Recorded Successfully!

Enter your choice : 8

=====

VOTING REPORT

=====

Pasta : 1 Votes

Enter your choice : 9

```
=====
                        HOSTEL MESS ANALYTICS REPORT
=====
Total Feedback Entries : 1
Average Taste Score : 3.0
Average Quantity Score : 5.0
Average Hygiene Score : 4.0
Average Presentation Score : 5.0
Total Complaints : 1
Total Waste Records : 1
Student Satisfaction Rate : 0.0 %
Overall Mess Rating : 4.25
Status : EXCELLENT
=====
```

Enter your choice : 10

Thank You For Using The System!

11. TESTING APPROACH

The project was tested using different input combinations to verify correct functionality.

Test Case	Expected Result	Status
Submit Feedback	Feedback stored successfully	Pass
Submit Complaint	Complaint recorded successfully	Pass
Add Waste Record	Waste data recorded	Pass
Vote for Dish	Vote counted properly	Pass
Generate Analytics	Reports displayed correctly	Pass

The testing demonstrated that all major functionalities worked as intended.

12. CHALLENGES FACED

- Designing meaningful analytics reports.
 - Organizing multiple modules effectively.
 - Ensuring proper integration between modules.
 - Creating a user-friendly interface.
 - Generating comprehensive reports using basic Python concepts.
-

13. LEARNINGS AND KEY TAKEAWAYS

The project provided valuable experience in:

- Python programming
- Modular application development
- Data processing and analysis
- Software design principles
- Problem-solving skills
- Report generation
- User-centric application development

The project also improved understanding of software development life cycle concepts.

14. FUTURE ENHANCEMENTS

The following improvements can be implemented in future versions:

- Database integration
- Web-based interface
- Mobile application
- AI-based recommendation system
- Automated report generation

- Graphical data visualization
 - Real-time notification system
-

15. REFERENCES

1. Python Documentation – <https://docs.python.org>
2. VIT Course Materials
3. Python Programming Lecture Notes